

Design and Analysis of a Split-Plot Experiment to Assess the Effects of Educational Intervention and Support Strategies on Adolescent Smoking

Michelle Navarrete Vega

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ABSTRACT

Background: Adolescent smoking is a major public health concern, with approximately 90% of smokers initiating before age 18. School-based interventions represent critical prevention opportunities. **Methods:** A split-plot ANOVA design was used across 11 districts. Schools were randomized to educational intervention (no presentation vs. two-hour testimonial presentation); homerooms within schools were randomized to one of four support strategies (none, video, ex-smoker Q&A, or both). The outcome was average cigarettes smoked per student in the prior week. **Results:** Support strategy was highly significant ($F(3, 60) = 38.99, P < 0.001$); strategies involving direct ex-smoker contact were associated with substantially lower smoking rates. The educational intervention alone did not reach significance ($P = 0.129$). **Conclusions:** Direct contact with a former smoker is the active ingredient driving smoking reduction. Future programs should prioritize the combined presentation and ex-smoker support strategy.

Keywords: adolescent smoking · split-plot design · school-based intervention · ANOVA · public health

1. Background

Adolescent smoking is a major public health concern because smoking during this developmental period is associated with impaired lung development, increased risk of respiratory infection, and elevated risk of lung cancer (2). Approximately 90% of smokers begin before the age of 18, and earlier initiation substantially increases the difficulty of cessation later in life (2). School-based interventions are therefore important targets for prevention efforts.

This study evaluated the effects of school-based educational and support interventions on cigarette smoking among eighth-grade students. The educational intervention consisted of either no presentation or a two-hour presentation featuring testimonials from former smokers. The support strategy consisted of one of four approaches: no support, a follow-up video, a question-and-answer session with a former smoker, or both combined.

The primary objective was to determine whether educational intervention and support strategy affected average cigarette consumption.

2. Methods

2.1 Study Design

A split-plot experimental design was used, with districts serving as blocks (1). Based on *a priori* power calculations, a minimum of 7 districts were required to achieve at least 90% power for both hypothesis tests at $\alpha = 0.05$; 11 districts were recruited.

Within each district, two schools were randomly assigned to one of two educational intervention levels. Within each school, four homerooms (8–12 students each) were randomly assigned to one of four support strategies. The educational intervention was the whole-plot factor; support strategy was the subplot factor. The response was the average number of cigarettes smoked per student during the week preceding a survey administered one month after treatment.

2.2 Hypotheses

Let α_i denote the effect of educational intervention I and γ_j the effect of support strategy J . The hypotheses were:

$$H_0: \alpha_1 = \alpha_2 = 0 \quad (\text{no educational intervention effect})$$

$$H_0: \gamma_1 = \gamma_2 = \gamma_3 = \gamma_4 = 0 \quad (\text{no support effect})$$

2.3 Statistical Analysis

A split-plot ANOVA was fitted with fixed effects for educational intervention, support strategy, and their interaction; district was treated as a random blocking factor:

$$Y_{ijk} = \mu + \alpha_i + \gamma_j + (\alpha\gamma)_{ij} + b_k + \varepsilon_{ijk} \quad (1)$$

where Y_{ijk} is the average cigarettes smoked per student for the J th support strategy within the I th educational intervention in the K th district; μ is the overall mean; α_i is the fixed effect of educational intervention; γ_j is the fixed effect of support strategy; $(\alpha\gamma)_{ij}$ is their interaction; b_k is the random district effect; and ε_{ijk} is the subplot residual. Educational intervention effects were tested against the district-by-intervention mean

square; subplot effects against the subplot residual. Ninety-five percent confidence intervals were constructed using the appropriate error term for each stratum. Pairwise comparisons among support strategies identified which strategies differed significantly.

All analyses were performed in R (v4.4.0) using `readxl`, `dplyr`, `ggplot2`, and `tidyverse`.

3. Results

3.1 Descriptive Statistics

Prior to hypothesis testing, descriptive statistics were examined across treatment groups (Table 1). Across both intervention groups, students in the Ex-Smoker and Video + Ex-Smoker conditions had lower average smoking rates than those receiving No Support or Video alone. Students who received the presentation had lower average rates across all support strategies.

Table 1. Average cigarettes smoked per student by treatment group; mean (SD).

Support	No Pres.	Pres.	Marginal
None	5.24 (1.88)	4.29 (1.75)	4.77
Video	5.94 (1.94)	4.75 (1.77)	5.34
Ex-Smoker	3.13 (0.93)	2.31 (1.07)	2.72
Video + Ex-Smoker	3.00 (1.41)	1.97 (0.99)	2.49
Marginal	4.33	3.33	

Homerooms in the No Presentation condition showed consistently higher median smoking rates across all support strategies (Figure 1). The interaction plot (Figure 2) showed nearly parallel lines, suggesting little evidence of an educational intervention $\times$ support strategy interaction.

3.2 Between-District Variability

Median district smoking rates ranged from approximately 2.3 (District 9) to 4.0 (Districts 1 and 2), confirming that blocking on district was appropriate (Figure 3). Diagnostic plots (Figure 4) showed residuals closely following the Q-Q reference line with no systematic pattern in residuals vs. fitted values, supporting normality and homoscedasticity. Three observations (26, 58, 61) exceeded the $4/N \approx 0.045$ Cook's distance threshold (values ≈ 0.07) but remained far below 1 and were retained.

3.3 Split-Plot ANOVA

Results are presented in Table 2. The interaction between educational intervention and support strategy was not significant ($F(3, 60) = 0.114$, $P = 0.952$). The educational intervention was not statistically significant ($F(1, 10) = 2.733$, $P = 0.129$). Support strategy was highly significant ($F(3, 60) = 38.988$, $P < 0.001$).

Table 2. Split-plot ANOVA for average cigarettes smoked per student.

Stratum	Source	df	SS	MS	F	p
Block	District	10	34.51	3.45	—	—
Whole Plot	Educ. Int.	1	21.97	21.97	2.733	0.129
	Residual	10	80.39	8.04		
Subplot	Support	3	136.61	45.54	38.988	<0.001
	Educ. $\times$ Supp	3	0.40	0.13	0.114	0.952
	Residual	60	70.08	1.17		

3.4 Confidence Intervals and Pairwise Comparisons

For educational intervention, both 95% CIs contained zero, consistent with the non-significant F -test (Table 3). For support strategy, CIs were narrower (subplot error, $df = 60$): the None and Video conditions had CIs entirely above zero, while Ex-Smoker and Video + Ex-Smoker had CIs entirely below zero, confirming their association with lower smoking rates.

Table 3. Point estimates and 95% CIs for treatment effects (sum-to-zero parametrization).

Effect	Estimate	95% CI
<i>Educational Intervention (whole-plot error, $df = 10$)</i>		
No Presentation ($\hat{\alpha}_1$)	0.499	(−0.174, 1.173)
Presentation ($\hat{\alpha}_2$)	−0.499	(−1.173, 0.174)
<i>Support Strategy (subplot error, $df = 60$)</i>		
None ($\hat{\gamma}_1$)	0.938	(0.539, 1.337)
Video ($\hat{\gamma}_2$)	1.515	(1.116, 1.914)
Ex-Smoker ($\hat{\gamma}_3$)	−1.111	(−1.510, −0.712)
Video + Ex-Smoker ($\hat{\gamma}_4$)	−1.342	(−1.741, −0.943)

Pairwise comparisons (Table 4, Figure 5) showed that the two within-group comparisons — None vs. Video (−0.576, 95% CI: −1.228 to 0.076) and Ex-Smoker vs. Video + Ex-Smoker (0.231, 95% CI: −0.421 to 0.883) — were not statistically significant. All four cross-group comparisons (ex-smoker contact vs. no contact) were significant, with differences ranging from 2.049 to 2.857 cigarettes per student.

Table 4. Pairwise differences in average cigarettes smoked between support strategies (subplot error, $df = 60$).

Comparison	Diff.	95% CI
None vs. Video	−0.576	(−1.228, 0.076)
None vs. Ex-Smoker	2.049	(1.397, 2.701)
None vs. Video+Ex-Smoker	2.280	(1.629, 2.932)
Video vs. Ex-Smoker	2.626	(1.974, 3.277)
Video vs. Video+Ex-Smoker	2.857	(2.205, 3.508)
Ex-Smoker vs. Video+Ex-Smoker	0.231	(−0.421, 0.883)

CIs not containing zero indicate $P < 0.05$.

4. Discussion

Support strategy was the primary driver of reduced cigarette consumption. Direct contact with a former smoker — either alone or combined with video — was associated with substantially lower average smoking rates compared to no support or video alone, identifying the interpersonal ex-smoker element as the active ingredient. Video alone showed no meaningful benefit over no support, suggesting passive media-based interventions are insufficient without an interpersonal component.

Although the educational presentation did not reach statistical significance ($P = 0.129$), the consistently lower rates in the Presentation group across all support conditions suggest a directional benefit. The non-significant result may partly reflect limited power at the whole-plot level, where only 11 districts contributed to the test. The recommended combination for future implementation is the two-hour presentation paired with the Video + Ex-Smoker support strategy, which had the largest estimated reduction ($\hat{\gamma}_4 = -1.342$).

4.1 Limitations

Individual-level smoking data were unavailable due to privacy constraints; homeroom-level averaging may have reduced sensitivity and obscured individual variation. Self-reported smoking data are subject to social desirability bias. Generalizability depends on whether the 11 recruited districts are representative of broader populations. Future studies with more districts and individual-level data would strengthen both power and precision.

References

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- [2] Park SH. Smoking and adolescent health. *Korean Journal of Pediatrics*. 2011;54(10):401–404. <https://doi.org/10.3345/kjp.2011.54.10.401>

Figures

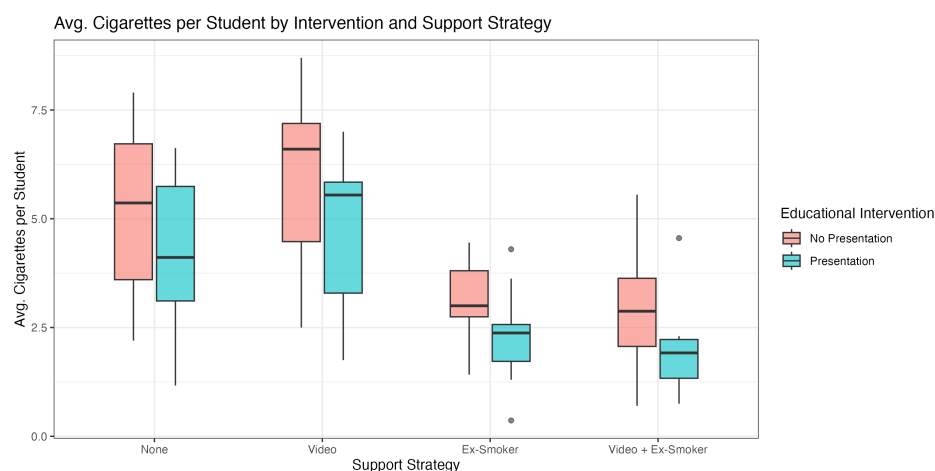


Figure 1. Boxplots of average cigarettes smoked per student by support strategy and educational intervention. Homerooms assigned to the No Presentation condition showed consistently higher median smoking rates across all four support strategies.

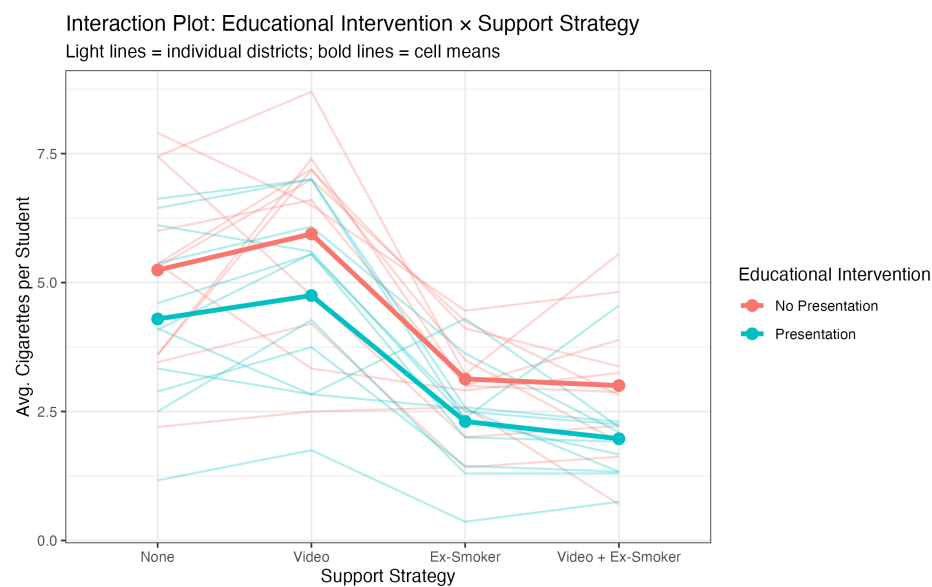


Figure 2. Interaction plot: average cigarettes smoked per student by support strategy and educational intervention. Nearly parallel lines indicate little evidence of an interaction between the two factors.

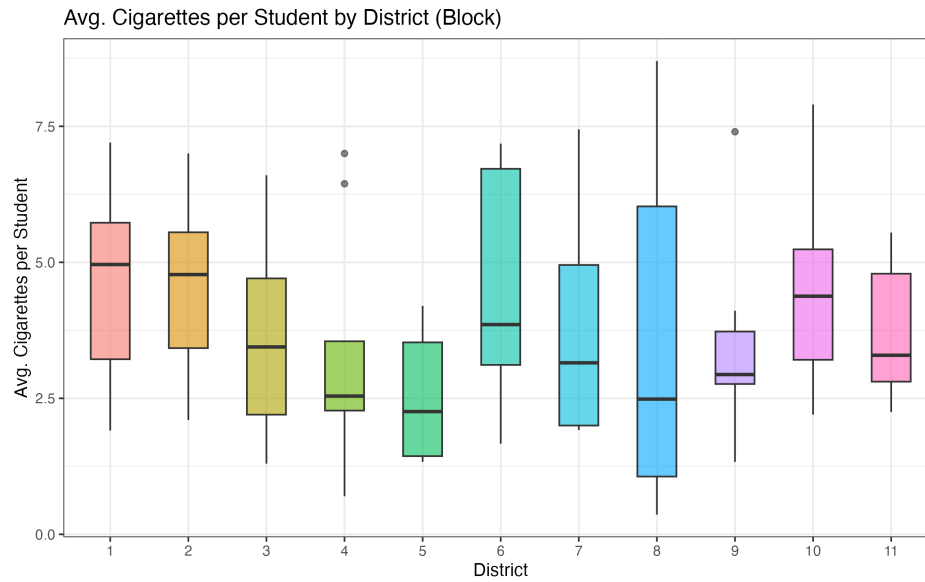


Figure 3. Boxplots of average cigarettes smoked per student by district. Considerable between-district variability (median ≈ 2.3 – 4.0) confirms that blocking on district was appropriate.

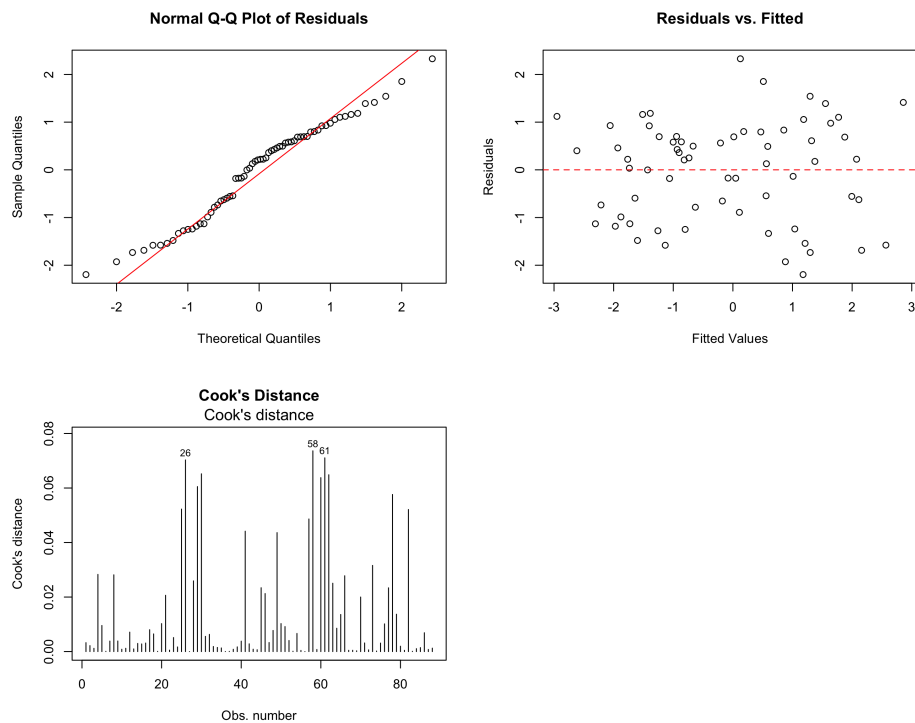


Figure 4. Model diagnostic plots. Normal Q-Q plot, residuals vs. fitted values, and Cook's distance. No serious violations of model assumptions were detected.

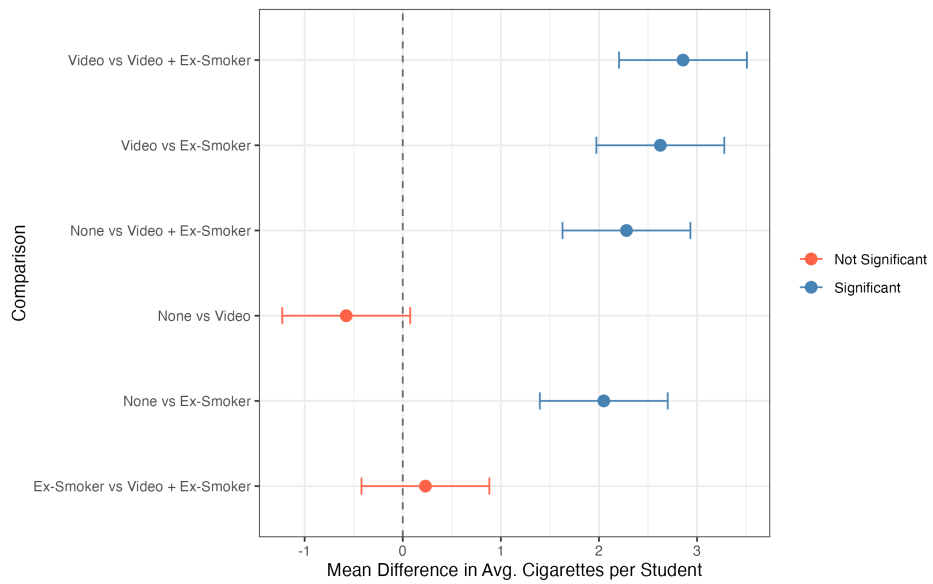


Figure 5. Pairwise differences in average cigarettes smoked per student between support strategies with 95% confidence intervals. Intervals not crossing zero indicate a statistically significant difference at $\alpha = 0.05$.